Homework Assignment 2

Math 311, Fall 2026

Rutgers University

Exercise 1.4.4 Let $a < b$ be real numbers, and consider the set $T = \mathbf{Q} \cap [a, b]$. Suppose $x \in T$. Then, $x \leq b$. So b is an upper bound for T. Suppose, for contradiction, that there exists some upper bound $c \in T$ such that

$$a < c < b.$$

By density of $\mathbb{Q}$ in $\mathbb{R}$, there exists some $q \in \mathbb{Q}$ such that

$$a < c < r < b.$$

Then

$$r \in \mathbb{Q} \cap [a, b] = T.$$

This contradicts the assumption that c is an upper bound on c. Thus c cannot be an upper bound on T. Therefore, b must be the least upper bound, or

$$b = \sup T.$$

Exercise 1.4.5 Let $a, b \in \mathbb{R}$ such that

$$a < b$$

So

$$a - \sqrt{2} < b - \sqrt{2}$$

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $r \in \mathbb{Q}$ such that

$$a - \sqrt{2} < r < b - \sqrt{2}$$

Then

$$a < r + \sqrt{2} < b.$$

Let $t = r + \sqrt{2}$. Since $r \in \mathbb{Q}$ and $\sqrt{2} \in \mathbb{I}$, by part (b) of 1.4.1, we have that $t \in \mathbb{I}$. Hence we have $t \in I$ where

$$a < t < b.$$

Exercise 1.4.8

(a) Take

$$A = (0,1) \cap \mathbb{Q}, \qquad B = (0,1) \setminus \mathbb{Q}.$$

Then $A \cap B = \emptyset$, and

$$\sup A = \sup B = 1 \notin A, B.$$

(b) Take

$$J_n = \left(-\frac{1}{n}, \frac{1}{n}\right) : n \in \mathbb{N}$$

Then,

$$J_1 \supseteq J_2 \supseteq J_3 \supseteq \dots.$$

Furthermore

$$\bigcap_{n=1}^{\infty} J_n = \{0\}.$$

Which non-empty and contains a finite number of elements, one in this case.

(c) Take

$$L_n = [n, \infty]$$

where $n \in \mathbb{N}$. Then

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \dots$$

Suppose, for contradiction, that

$$\bigcap_{n=1}^{\infty} L_n \neq \emptyset.$$

Then, there exists a $x \in L_n$ such that

$$x \geq n$$

for every $n \in \mathbb{N}$. But, by Archimedean Property, there exists a $n > x$ for every single x. Thus, this is a contradiction and no such x exists. Therefore,

$$\bigcap_{n=1}^{\infty} L_n = \emptyset.$$

(d) We can use the same interval from part (c), that being

$$I_n = [n, \infty].$$

For any finite N,

$$\bigcap_{n=1}^{N} I_n = [N, \infty] \neq \emptyset.$$

But as you take $N \to \infty$,

$$\bigcap_{n=1}^{\infty} I_n = \emptyset.$$

Exercise 1.5.9

(a) Let $x = \sqrt{2}$. Then

$$x^2 - 2 = 0.$$

So $\sqrt{2}$ is algebraic

Now let $x = \sqrt[3]{2}$. Then

$$x^3 - 2 = 0.$$

So $\sqrt[3]{2}$ is algebraic as well.

Finally let $x = \sqrt{3} + \sqrt{2}$. Then

$$x^4 - 10x^2 + 1 = 0.$$

So $\sqrt{3} + \sqrt{2}$ is algebraic.

(b) Fix $n \in \mathbb{N}$, and let A_n be the algebraic numbers obtained as roots of polynomials with integer coefficients that have degree n. Recall that every polynomial has a finite number of roots.

We can represent every degree-n polynomial by a finite sequence of integers like

$$(a_0, a_1, ..., a_n) \in \mathbb{Z}^{n+1}$$

Since $\mathbb{Z}$ is countable, the finite cartesian product $\mathbb{Z}^{n+1}$ is countable as well. Hence there are only countably many degree-n polynomials with integer coefficients.

Each such polynomial has finitely many roots, so A_n is a countable union of finite sets. Therefore A_n is countable.

(c) Let A be the set of all algebraic numbers. Since every algebraic number is a root of some polynomial of finite degree, it follows that

$$A = \bigcup_{n=1}^{\infty} A_n.$$

From part (b) we have that A_n is countable, and so A is the countable union of countable sets. Therefore A is countable.

Furthermore, since

$$\mathbb{R} = A \cup T$$

and $\mathbb{R}$ is uncountable, it follows that T is uncountable as well. Otherwise, $\mathbb{R}$ would be the union of two countable sets and countable, which is a contradiction.

Exercise 1.5.11

(a) Suppose you partition

$$X = A \cup A', \qquad Y = B \cup B'$$

with

$$A \cap A' = \varnothing, \qquad B \cap B' = \varnothing,$$

such that f maps A onto B and g maps B' onto A'. Then define $h : X \to Y$ by

$$h(x) = \begin{cases} f(x), & x \in A, \\ g^{-1}(x), & x \in A' \end{cases}$$

Then you can show that h maps $A \to B$ bijectively and $A' \to B'$ bijectively. Hence

$$X \sim Y.$$

(b) Set

$$A_1 = X \setminus g(Y) = \{x \in X : x \notin g(Y)\},$$

and define

$$A_{n+1} = g(f(A_n)).$$

Then

$$A_{n+1} \subseteq g(Y)$$

for every $n \geq 1$, and

$$A_1 \cap g(Y) = \emptyset.$$

So

$$A_n \cap A_n = \emptyset$$

for all $n \geq 2$.

Suppose, for contradiction, that

$$A_m \cap A_n \neq \emptyset$$

for some $m < n$.

Since

$$A_m = g(f(A_{m-1})), \qquad A_n = g(f(A_{n-1})),$$

and both f and g are injective, it follows that

$$A_{m-1} \cap A_{n-1} \neq \emptyset.$$

Repeating this inductively gives us

$$A_1 \cap A_{n-1} \neq \emptyset.$$

But this contradicts that $A_1 \cap A_n = \emptyset$. Thus each A_n is pairwise disjoint from each other. Since f is injective, their images are also pairwise disjoint.

(c) Let

$$A = \bigcup_{n=1}^{\infty} A_n, \qquad B = \bigcup_{n=1}^{\infty} f(A_n).$$

Then

$$f(A) = f\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} f(A_n) = B.$$

Since f is injective on X and $A \subseteq X$, f is also injective on A. Hence $f : A \to B$ is a bijection.

(d) Let

$$A' = X \setminus A, \qquad B' = Y \setminus B.$$

Take $y \in B'$. Suppose, for contradiction, $g(y) \in A$. Then $g(y) \in A_n$ for some $n \geq 2$. So

$$g(y) = g(f(A_{n-1})).$$

Thus $g(y) = g(f(x))$ for some $x \in A_{n-1}$. Since g is injective,

$$y = f(x) \in f(A_{n-1}) \subseteq B,$$

contradicting $y \in B'$. Therefore $g(y) \in A'$, so $g(B') \subseteq A'$. Now take $x \in A'$. Since $x \notin A_1$, we have $x \in g(Y)$. Thus there exists $y \in Y$ such that

$$g(y) = x.$$

Suppose, for contradiction, $y \in B$. Then $y \in f(A_n)$ for some n, so $y = f(z)$ for some $z \in A_n$. Hence

$$x = g(y) = g(f(z)) \in A_{n+1} \subseteq A,$$

contradicting $x \in A'$. Therefore $y \notin B$, so $y \in B'$. Hence every $x \in A'$ is hit by some $y \in B'$, so g maps B' onto A'.

Exercise 1.6.1 If $(0, 1)$ is uncountable, then $\mathbb{R}$ is uncountable as well because

$$(0, 1) \subseteq \mathbb{R}.$$

Suppose $\mathbb{R}$ is uncountable. Define $f : (0, 1) \to (-1, 1)$ by

$$f(x) = 2x - 1.$$

Suppose

$$f(x_1) = f(x_1).$$

Then

$$2_{x1} - 1 = 2_{x_2} - 1,$$

so

$$x_1 = x_2$$

Let $y \in (-1, 1)$ Choose

$$x = \frac{y + 1}{2}.$$

Since $-1 < y < 1$,

$$0 < \frac{y + 1}{2} < 1,$$

so $x \in (0, 1)$. Also,

$$f(x) = 2\left(\frac{y + 1}{2}\right) - 1 = y.$$

Thus f is bijective. From example 1.5.4. we have the bijection $g : (-1, 1) \to \mathbb{R}$ defined by

$$g(x) = \frac{x}{x^2 - 1}.$$

Thus

$$g \circ f : (0, 1) \to \mathbb{R}$$

is a bijection. Therefore,

$$(0, 1) \sim \mathbb{R}.$$

Exercise 1.6.9 Define $f : \mathcal{P}(\mathbb{N}) \to \mathbb{R}$ by

$$f(S) = \sum_{n \in S} 10^{-n}.$$

Suppose $f(S_1) = f(S_2)$. The decimal expansion of $f(S_i)$ has a 1 in the nth decimal place if and only if $n \in S_i$, and a 0 otherwise.

Since $f(S_1) = f(S_2)$, their decimal expansions must agree in every decimal place by defition of f. That is,

$$n \in S_1 \iff n \in S_2$$

for every $n \in \mathbb{N}$. Hence $S_1 = S_2$. Therefore, f is injective.

For each $x \in \mathbb{R}$ define

$$S_x = \{q \in \mathbb{Q} : q < x\}.$$

Then

$$S_x \subseteq \mathbb{Q}.$$

If $x \neq y$, say $x < y$ WLOG, then by density of $\mathbb{Q}$, there exists $q \in \mathbb{Q}$ such that

$$x < q < y.$$

So

$$q \notin S_x \qquad \text{but} \qquad q \in S_Y$$

Thus $S_x \neq S_y$. Hence $x \to S_x$ is injection for $\mathbb{R} \to \mathcal{P}(\mathbb{Q})$. Recall that $\mathbb{Q} \sim \mathbb{N}$, so $\mathcal{P}(\mathbb{Q}) \sim \mathcal{P}(\mathbb{N})$

Thus, by Schroder-Bernstein theorem, we conclude that there exists a bijection between $\mathbb{P}(\mathbb{N})$ and $\mathbb{R}$. Therefore,

$$\mathbb{P}(\mathbb{N}) \sim \mathbb{R}.$$

Exercise 1.6.10

(a) Let $\mathcal{F}$ be the set of all functions from $\{0, 1\} \to \mathbb{N}$. Define $\Phi : \mathcal{F} \to \mathbb{N} \times \mathbb{N}$ by

$$\Phi(f) = (f(0), f(1)).$$

Suppose $\Phi(f) = \Phi(g)$. Then

$$(f(0), f(1)) = (g(0), g(1)),$$

so

$$f(0) = g(0), \qquad f(1) = g(1).$$

Since the domain only contains 0 and 1, this means f and g agree at every point in their domain. Therefore

$$f = g.$$

So Φ is injective.

Take

$$(m, n) \in \mathbb{N} \times \mathbb{N}.$$

Define a function

$$f : \{0, 1\} \to \mathbb{N}$$

by

$$f(0) = m, \qquad f(1) = n.$$

Then

$$\Phi(f) = (m, n).$$

Thus Φ is surjective, and so Φ is bijective as well. So

$$\mathcal{F} \sim \mathbb{N} \times \mathbb{N}$$

Since $\mathbb{N} \times \mathbb{N}$ is countable, it follows that $\mathcal{F}$ is countable.

(b) Let $\mathcal{G}$ be the set of all functions from $\mathbb{N} \to \{0, 1\}$. For each of these functions g, define

$$S_f = \{n \in \mathbb{N} : f(n) = 1\}.$$

Then $S_f \subseteq \mathbb{N}$, so $S_f \in \mathcal{P}(\mathbb{N})$. Conversely, given any $S \subseteq \mathbb{N}$, define

$$f_S(n) = \begin{cases} 1, & n \in S, \\ 0, & n \notin S. \end{cases}$$

Thus every function corresponds to exactly one subset of $\mathbb{N}$, and every subset corresponds to exactly one such function. Hence

$$\mathcal{G} \sim \mathcal{P}(\mathbb{N}).$$

Since $\mathcal{P}(\mathbb{N})$ is uncountable, it follows that $\mathcal{G}$ is uncountable.

(c) For each subset $S \subseteq \mathbb{N}$, define

$$A_S = \{2n : n \in S\} \cup \{2n + 1 : n \notin S\}.$$

Let

$$\mathcal{A} = \{A_S : S \subseteq \mathbb{N}\}.$$

Suppose $S \neq T$. Then there exists some n belonging to S or T but not the other. Say

$$n \in S, \qquad n \notin T.$$

Then

$$2n \in A_S \qquad \text{but} \qquad 2n \notin A_T,$$

so

$$A_S \not\subseteq A_T.$$

Furthermore,

$$2n+1 \in A_T \qquad \text{but} \qquad 2n+1 \notin A_S,$$

so

$$A_T \not\subseteq A_S.$$

Therefore no two distinct elements of $\mathcal{A}$ contain one another, so $\mathcal{A}$ is an antichain. Also, if $S \neq T$, then $A_S \neq A_T$, so the map

$$S \mapsto A_S$$

is injective. By definition of $\mathcal{A}$, it is also surjective onto $\mathcal{A}$. Therefore,

$$\mathcal{A} \sim \mathcal{P}(\mathbb{N}).$$

Since $\mathcal{P}(\mathbb{N})$ is uncountable, it follows that $\mathcal{P}(\mathbb{N})$ contains an uncountable antichain.